

Portfolio Game - Report 1

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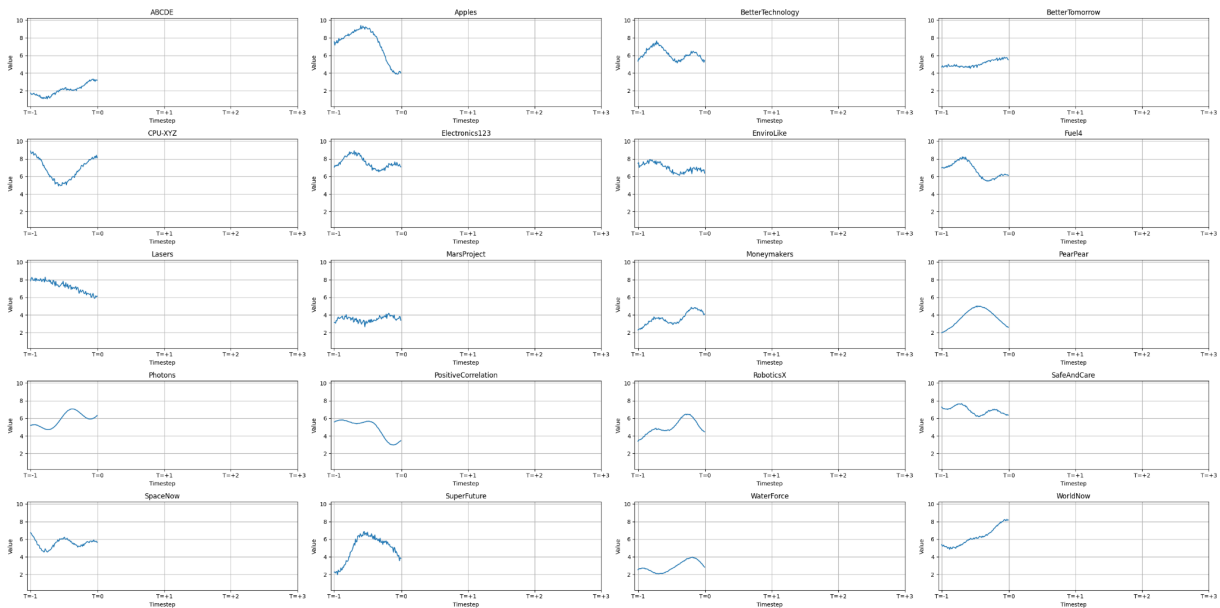
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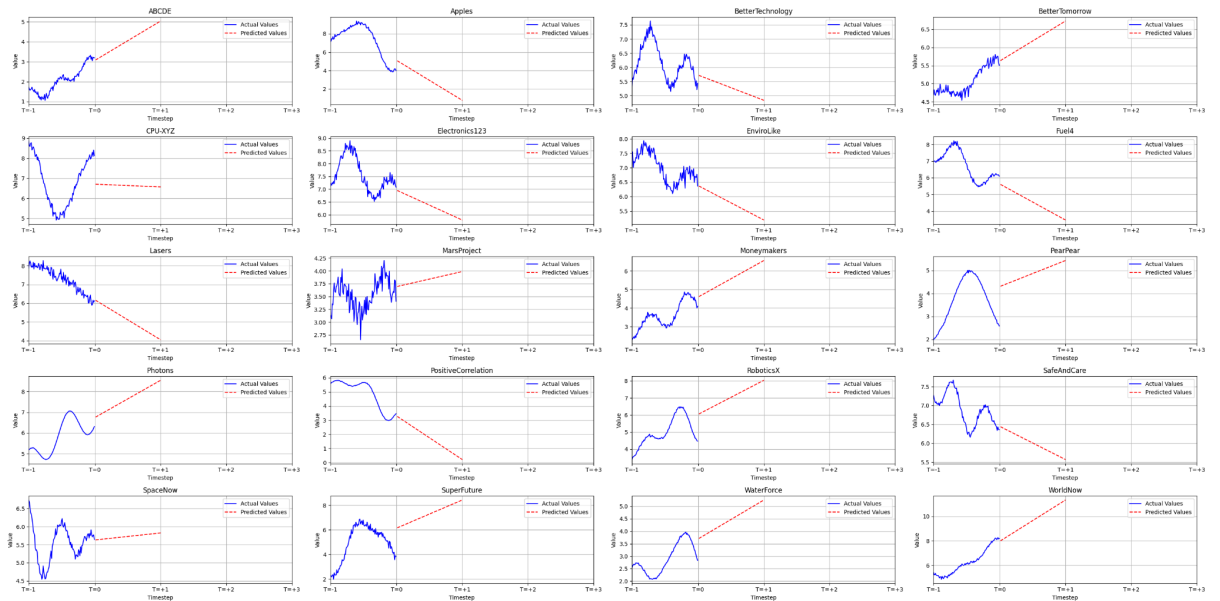
1. Predictions

The “Portfolio Game” is a project based on predicting stock prices for 20 fictitious companies and utilizing Markovitz to create a Pareto front of non-dominated solutions that balance exploration and exploitation. To achieve this task, a proper regression model must be implemented to predict companies' prices at future time points. As such, multiple models were tested. The following subsections showcase the results of different regression methods.

1.0 Original Data

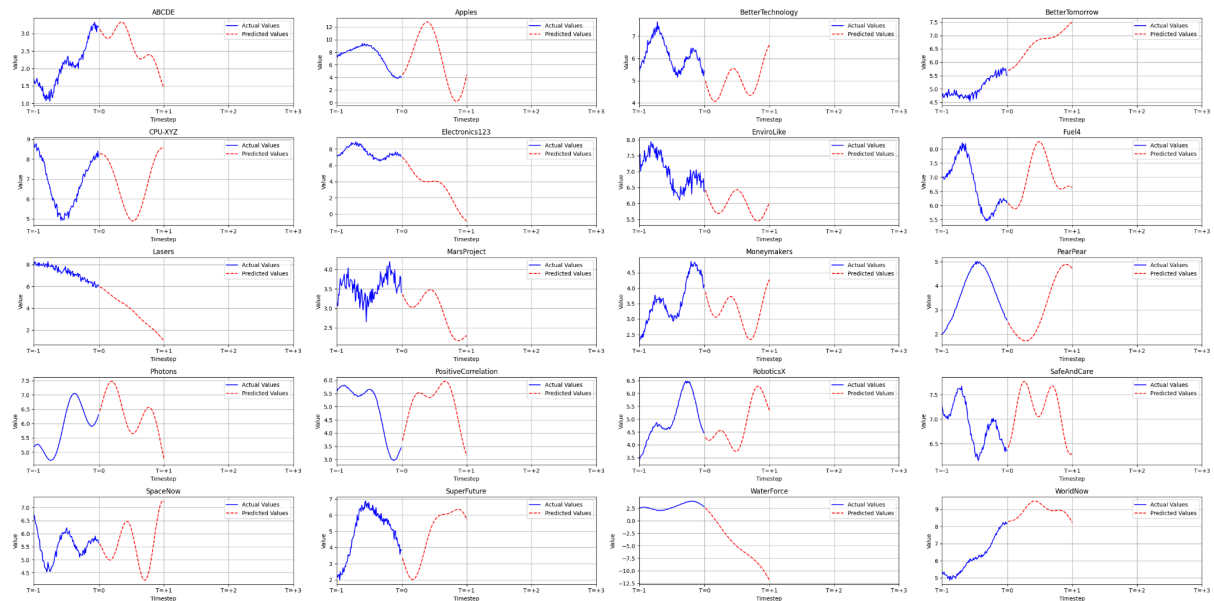


1.1 Linear Regression



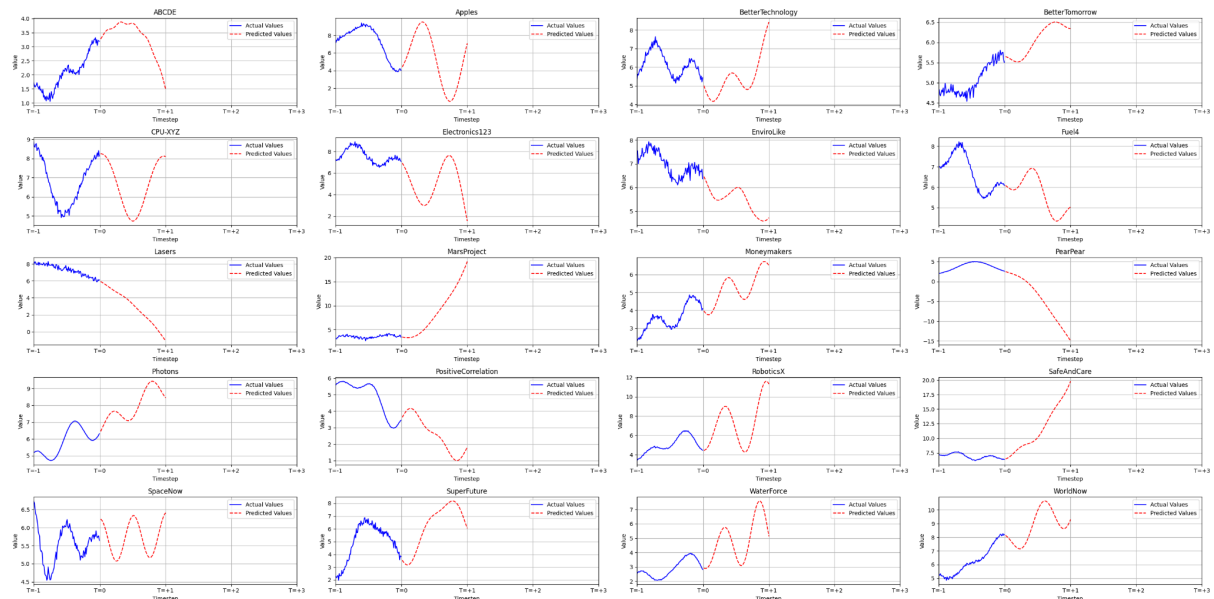
1.2 Sinusoidal Regression

This solution was based on fitting a curve defined as $A1 * \sin(B1 * x + C1) + A2 * \sin(B2 * x + C2) + D$ to the data.



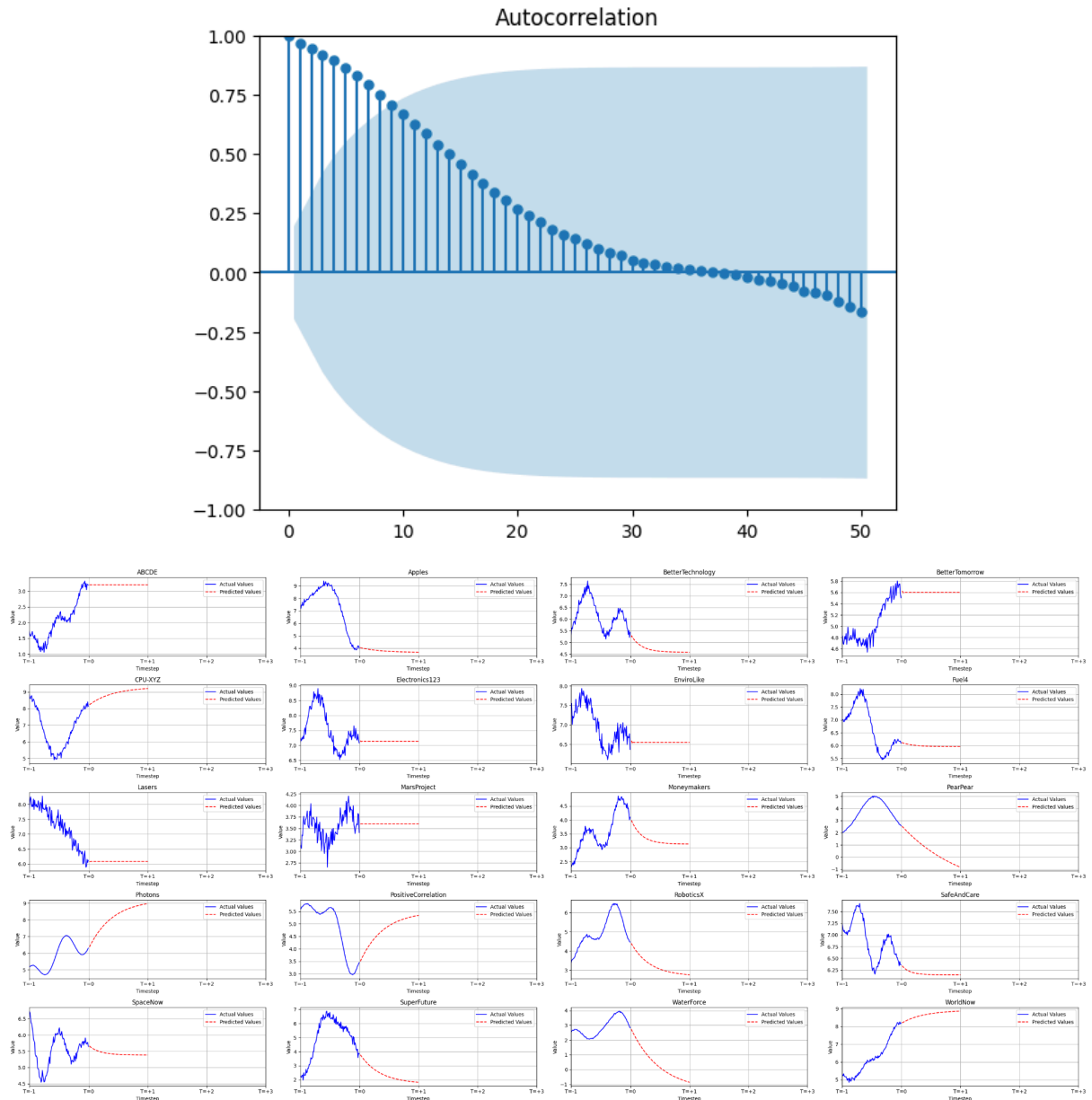
1.3 Complex Sinusoidal Regression

A more complex version of the previous solution, defined as fitting the curve $a * x + A1 * \sin(B1 * x + C1) + A2 * \sin(B2 * x + C2) + D$ to the data.

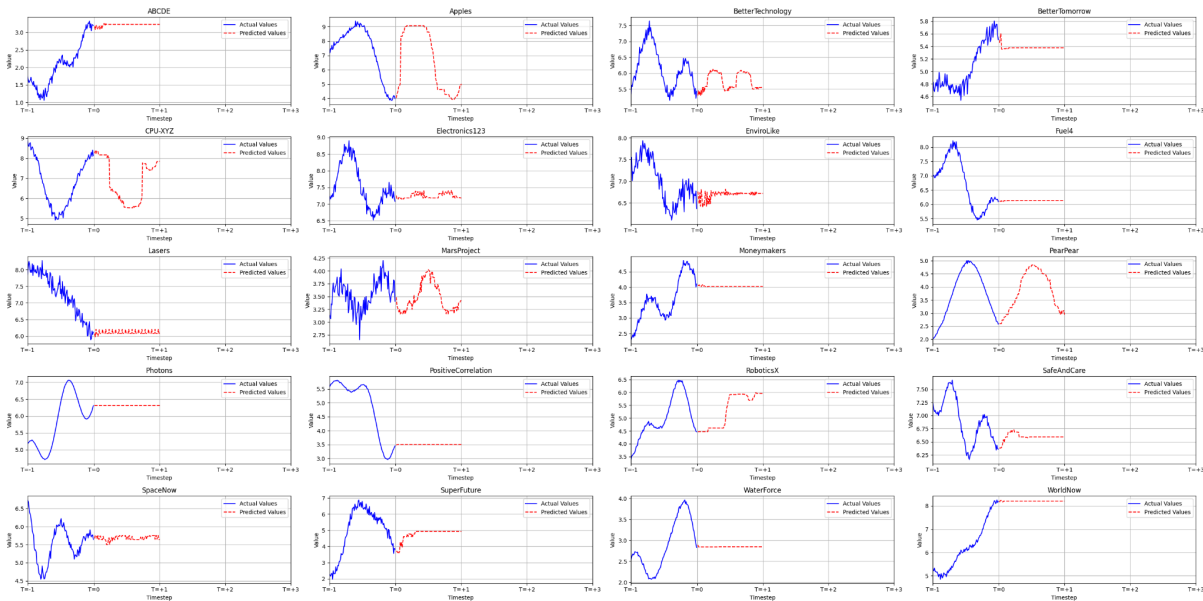


1.4 ARIMA

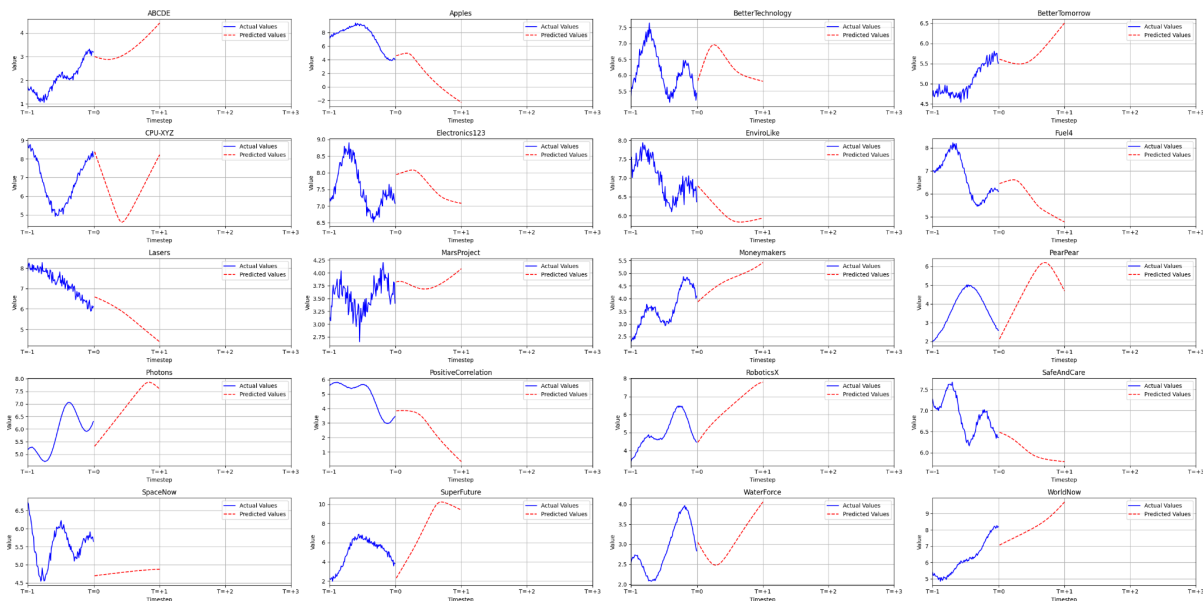
This solution is based on the well-known ARIMA method with the parameters being set by hand based on the autocorrelation and partial autocorrelation plots.



1.5 XGBoost



1.6 NODE

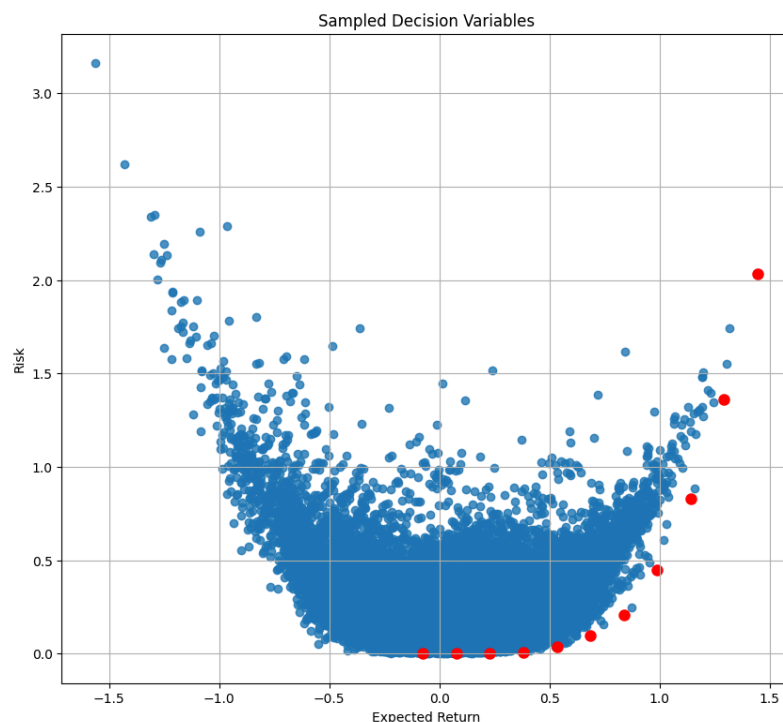


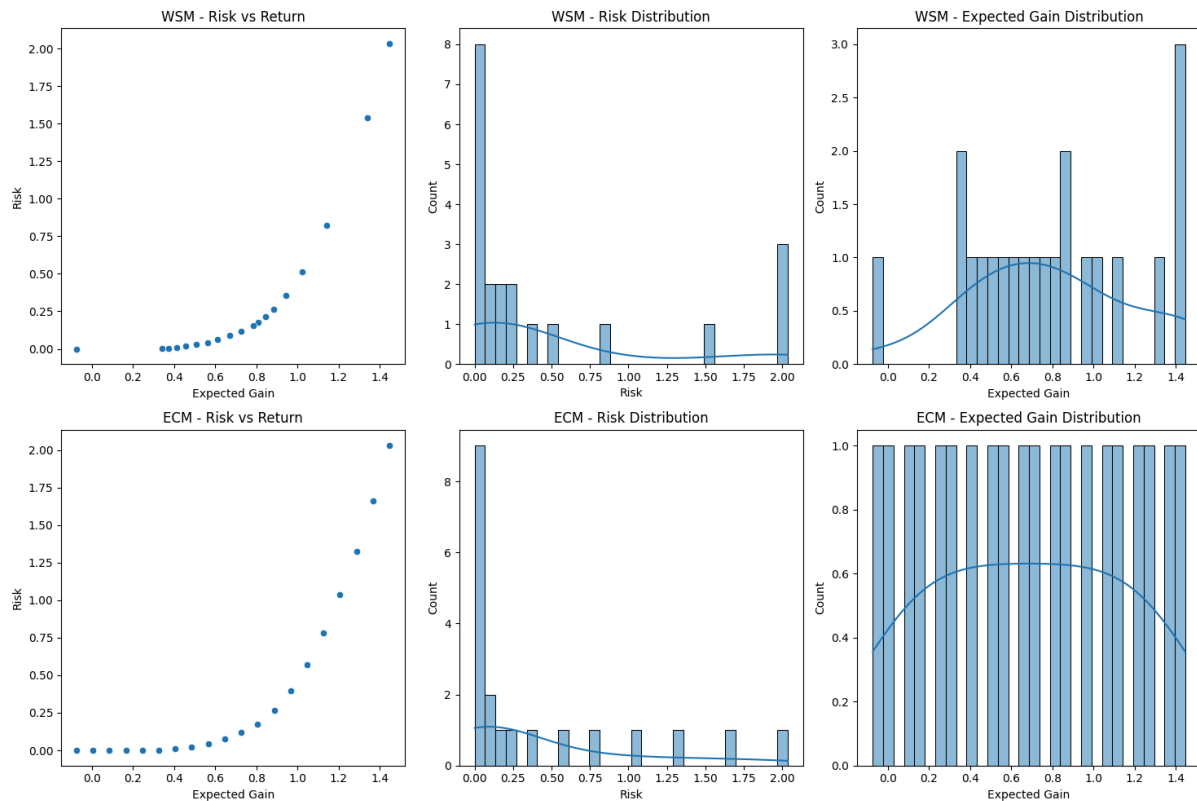
Conclusions

Many methods were either too simplistic to capture the true nature of the data (linear regression, ARIMA) or struggled with the small number of training examples (XGBoost, NODE). As such, no clear winner has appeared from the tests. Unfortunately, some of the most promising solutions like the sinusoidal regression functions ended up greatly surpassing the ranges of values produced by the original functions, potentially causing the model to be unstable. As such, we have decided to pick **NODE**'s predictions to base the solution on, as it seems to capture some trends well, not greatly overestimating or underestimating the potential future values.

2. Pareto Front

Two solutions were implemented by us as of the time of writing this report that allowed for finding the Pareto front - the WSM and ECM methods. In the end, the final solution utilized the ECM method as our experiments have showcased that it more uniformly samples the non-dominated solutions, providing a better overview of the available options. The below visualizations showcase a subset of non-dominated solutions for the game, as well as a comprehensive comparison between the ECM and WSM methods.





3. Solution

In the end, due to the difficulties of capturing the trends by the models, as well as, the fact that NODE risks overestimating the value for stocks such as “SuperFuture”, we have decided to choose the solution with a relatively small risk while providing a positive expected return. Especially, since due to the nature of the game, the most important aspect is to go into the green, rather than gather as much money as possible, choosing a safe option might be the best choice. However, diversifying the portfolio too much might risk not taking into account the regression at all. As such, we have identified the solution: **[111.7229029981 0.0005054091 0.0000000244 0.0000000086 0.0000001247 0.0542921372 0.1761429231 0.0234277488 0.1583375966 0.0343588227 0.0059541302 0.0405949427 0.0163776905 0.0000000706 0.1603273564 0.0108636238 0.1572035995 0.0000001091 0.0048063507 0.0000010368 0.0914914603 0.0658202431]** as the best candidate for this problem, due to its extremely small risk with a decent potential return. For a more thorough analysis, please refer to the project’s repository: <https://github.com/MatTheTab/MOO/tree/main>